

Beans Together

Concept Document

Tiago Pinto m68106

Pitch Line

In "Beans Together", four players cooperate on a checkers-inspired board where every bean counts: gathering more resources hinders movement, and a single mistake can lead the whole team to defeat.

Introduction

Beans Together is a cooperative board game for four players, focused on collective resource management, group decision-making, and a delicate balance between risk and progress. Players move across a checkerboard-inspired board, advancing each round while facing the unpredictable effects of cards and special spaces.

The core mechanic revolves around beans, a resource indirectly shared by all. Throughout the game, players accumulate beans through cards and special spaces, but this growth is a double-edged sword: **if any player reaches seven beans, the group loses immediately**. Victory is only achieved when the team manages to deplete the entire deck without any player exceeding this critical limit.

Each round introduces new challenges through cards with varied effects depending on their suit, as well as special spaces on the board—represented by brown and white checker pieces—that strategically alter the game state. Player movement is directly tied to their bean count: **the more beans they have, the fewer spaces they can move**, creating a cycle of complex decisions and progressive consequences.

In **Beans Together**, the goal is not to reach the end of a path, but to **survive until the last card of the deck is drawn**. The game encourages constant communication, joint planning, and cooperation, challenging the group to find the perfect balance between advancement, resource accumulation, and preventing collective collapse.

Demographic Breakdown

Beans Together is an inclusive game, suitable for people of any gender and age. Its rules are straightforward and easy to learn, yet the game allows for strategic depth. Furthermore, no computer is required to play.

Feature List

- 4-Player Cooperative Multiplayer;
- Checkers-Style Board Layout;
- Turn-Based Rounds with Card Draws;
- Suit-Based Card Effects;
- Special Board Tiles (White & Brown Checkers);
- Beans as the Core Shared Resource;
- Individual Bean Limit (7 = Team Loss);
- Dynamic Movement Based on Bean Count (More Beans = Less Movement);
- Collective Win/Loss Conditions;
- Deck Completion as Primary Objective;
- High Communication and Team Coordination;
- Strategic Resource Management under Pressure.

Feature List Breakdown

- **4-Player Cooperative Gameplay:** Four players work together as a team, sharing resources and making strategic decisions to achieve a common goal.
- **Checkers-Style Board Layout:** The game takes place on a grid-based board inspired by checkers, where positioning and movement routes are crucial for team success.
- **Turn-Based Rounds with Card Draws:** Gameplay is organized in turns, with players drawing cards each round.
- **Suit-Based Card Effects:** Each card has a suit with distinct effects, enabling diverse strategic approaches and player combinations.
- **Special Board Tiles (White & Brown Checkers):** Special tiles on the board trigger positive or negative effects, adding risk, variety, and situational decisions to movement.

- **Beans as a Shared Core Resource:** Beans function as the game's primary resource, managed collectively and used for actions, effects, and progression.
- **Individual Bean Limit (7 = Team Loss):** If any player accumulates more than 7 beans, the entire team loses immediately, emphasizing the importance of collective risk management.
- **Dynamic Movement Based on Bean Count:** A player's mobility decreases as they accumulate beans, creating a constant trade-off between power and positioning.
- **Collective Win and Loss Conditions:** The game is won or lost as a team, with no individual objectives, promoting full cooperation.
- **Deck Completion as the Primary Objective:** The main goal is to complete the card deck, requiring efficient management of resources, time, and positioning.
- **High Communication and Team Coordination:** Constant communication is essential for planning turns, sharing beans, and mitigating risks as a team.
- **Strategic Resource Management Under Pressure:** Players face critical decisions that demand a balance between progress, safety, and collective efficiency.

Formal Description

Game State:

The game state contains all relevant information at a given moment during the match:

- **players:** fixed set of 4 players: {Green, Yellow, Blue, Red};
- **current_player_index:** index of the current player (integer, 0–3);
- **player_positions:** each player's position on the circular board (dictionary String → int, values 0–63);
- **marks:** list of board space markings/states (array, size 64);
- **deck:** remaining card deck (array);
- **discard_pile:** cards already used (array);
- **beans:** number of beans each player holds (dictionary String → int);
- **is_moving:** indicates whether a piece is currently moving (boolean);
- **is_waiting_for_trade:** indicates whether the game is waiting for a trade decision (boolean);
- **trade_source_player:** player who initiated the trade (string).

S=

{players,current_player_index,player_positions,deck,discard_pile,beans,is_moving,is_waiting_for_trade}

Game View:

All players have access to the following information in real-time:

- Each player's bean count;
- The currently revealed card;
- The number of cards remaining in the deck;
- The position of all pieces on the board;
- Visual highlight indicating whose turn it is;
- The exchange menu (when active).

There is no hidden information among players. The only exception is the future order of the cards in the deck, which remains unknown to everyone.

Game Setup:

At the start of the game:

- All game pieces begin at position 0 on the board.
- All players start with 0 beans.
- The deck is created from the card files and shuffled.
- current_player_index = 0
- is_moving = false
- is_waiting_for_trade = false

Player Actions:

During their turn, a player may take the following actions:

Draw a Card:

- Reveal the top card of the deck and immediately apply its effect.

Move Their Piece:

- The player's piece automatically advances on the board.
- The number of spaces moved is determined by how many beans they have:
 - **0 – 3 beans:** move 3 spaces.

- **4 – 5 beans:** move 2 spaces.
- **6 or more beans:** move 1 space.

Transfer a Bean (conditional action):

- If the card effect allows it, the player may give 1 bean to any other player.

$A = \{\text{draw_card}(), \text{move}(\text{step}), \text{trade}(\text{target_player})\}$

Victory Conditions:

Lose Conditions:

The game ends immediately when any player reaches or exceeds the maximum number of beans.

- $\text{beans}[\text{player}] \geq \text{MAX_BEANS} (7)$

In this case, the result displayed is Defeat.

- $D = \exists p \in \text{players} | \text{beans}(p) \geq 7$

Victory:

If the deck runs out without any loss condition being met, the game ends in Victory.

$V = \text{deck} = \emptyset \wedge \forall p, \text{beans}(p) < \text{MAX_BEANS}$

Progression of Play:

The game flow proceeds as follows:

1. The current player draws a card.
2. The card's effect is applied.
3. The player's piece is moved automatically.
4. The effect of the landing space is applied.
5. If there is no pending turn swap:
 - The turn passes to the next player.
6. The game continues until a win or loss condition is met.